

Probing Quantum Nonlocality with Spacetime-Resolved $H \rightarrow \tau^+ \tau^-$ Decays

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We show that a future e^+e^- Higgs factory can perform a spacetime-resolved test of quantum nonlocality in Higgs boson decays. In $e^+e^- \rightarrow ZH \rightarrow \mu^+\mu^-\tau^+\tau^-$ events at $\sqrt{s} = 240$ GeV, the tau decay vertices are reconstructed with the impact-parameter method, giving event-by-event access to the spacetime interval between the two decays. Because the Higgs is a scalar, the $\tau^+\tau^-$ pair is produced in the maximally entangled triplet $(|\uparrow\uparrow\rangle + |\downarrow\downarrow\rangle)/\sqrt{2}$, with spin correlation matrix $C = \text{diag}(+1, +1, -1)$ exactly, for any tau velocity. We construct unbiased estimators of the CHSH score and of a concurrence witness, derive the exact small-sample noise bias of the Horodecki parameter, and build a Neyman–Pearson-optimal, permutation-calibrated test of the hypothesis that the correlations are carried by a signal of finite speed v_ψ . The causal-speed distribution is shown to follow an exact Pareto law, $P(v_{\text{sig}} > v) = \beta/v$, yielding the closed-form exclusion reach $v_\psi^{95} \simeq N\beta/(4z_{95}^2)$. In a simulated sample of 90 fully reconstructed $\pi^+\pi^-$ events—all with spacelike-separated decay vertices—entanglement is certified at 3.8σ and finite-speed causal models are excluded up to $v_\psi \approx 5c$, consistent with the analytic reach. Scaling to the full ILC program, the exclusion extends beyond 10^2c , providing the first test of quantum nonlocality in a Yukawa-mediated process with massive fermions.

The decay $H \rightarrow \tau^+\tau^-$ produces a maximally entangled pair of massive fermions. Because the Higgs boson is a spin-0 particle, the $\tau^+\tau^-$ system is created in a pure, maximally entangled spin state, regardless of the tau velocity [19–21]. Measuring the spin correlations of this system tests quantum mechanics in the relativistic regime, with direct sensitivity to Bell nonlocality [1–3].

Recent landmark measurements at the LHC, building on the proposal of Ref. [15], have established quantum entanglement in top quark pairs [16, 17] and provide evidence for entangled $H \rightarrow \tau^+\tau^-$ decays [18]. However, these measurements integrate over the spacetime structure of the decays: they demonstrate *that* the particles are entangled but do not probe *where* or *when* the entanglement is manifest. In photon-based Bell tests, the spacelike separation of measurements is enforced by the experimental design [4–6], and dedicated experiments bound the speed of any hypothetical superluminal influence [7]. At a collider, the tau leptons propagate freely and decay at spacetime positions governed by quantum mechanics; the spacelike separation is not imposed but *measured*.

There is also a historical motivation for testing quantum correlations specifically in weak-sector processes. The weak interaction has repeatedly overturned assumptions about fundamental symmetries: parity [9, 10], CP [11], and lepton-flavor conservation [12] were each found to be violated when tested. Quantum nonlocality has been verified extensively with photons [4–6] but never directly tested in an interaction mediated by the Higgs field. The Yukawa coupling $y_\tau H \bar{\tau}\tau$ is the only known interaction that produces maximally entangled massive-fermion pairs from a scalar source. A direct test of nonlocality in this channel is therefore well motivated, inde-

pendent of theoretical prejudice.

In this Letter we demonstrate that a future e^+e^- Higgs factory such as the ILC [28] enables precisely this measurement. The enabling capabilities are: (i) the known beam four-momentum, which fixes the Higgs rest frame through the Z recoil; (ii) the impact-parameter method of Jeans [26], which reconstructs the tau decay vertex from charged-pion track geometry; and (iii) the micron-level vertex resolution of detectors such as the ILD [29]. We further develop the statistical framework required for a defensible claim: exact Standard-Model (SM) expectations, unbiased estimators with closed-form variances, an optimal likelihood-ratio (LR) test calibrated by permutation, and an analytic sensitivity law that is validated on simulation.

Exact SM expectation.—For the decay of a scalar to two spin- $\frac{1}{2}$ fermions, the spin density matrix of the $\tau^+\tau^-$ pair is

$$\rho = \frac{1}{4} [\mathcal{K} \otimes \mathcal{K} + B_i^+ \sigma_i \otimes \mathcal{K} + B_j^- \mathcal{K} \otimes \sigma_j + C_{ij} \sigma_i \otimes \sigma_j], \quad (1)$$

with $i, j \in \{n, r, k\}$ the orthonormal basis built from the τ^- flight direction $\hat{k}$ in the Higgs rest frame, the transverse beam component $\hat{r}$, and $\hat{n} = \hat{k} \times \hat{r}$. A CP-even scalar coupling $\bar{\tau}\tau$ admits a single partial wave: $J = 0$ with parity $(-1)^{L+1} = +1$ forces $L = S = 1$, the 3P_0 wave. Along the decay axis only $m_L = 0$ survives, so the spin state is *exactly* the triplet $|\psi^+\rangle = (|\uparrow\uparrow\rangle + |\downarrow\downarrow\rangle)/\sqrt{2}$, giving $B_i^\pm = 0$ and

$$C_{\text{SM}} = \text{diag}(+1, +1, -1), \quad (2)$$

independent of the tau velocity β . More generally, for a CP-mixed coupling $\bar{\tau}(\cos\Delta + i\gamma_5 \sin\Delta)\tau$ the transverse block rotates, $C_{nn} = C_{rr} = \cos 2\Delta$, $C_{nr} = -C_{rn} =$

$\sin 2\Delta$, $C_{kk} = -1$, so that $C^T C = \mathbb{K}$ and the state remains maximally entangled for any Δ : the Horodecki parameter [13]—the sum m_{12} of the two largest eigenvalues of $C^T C$ —equals 2, the concurrence equals 1, and the maximal CHSH score is $2\sqrt{m_{12}} = 2\sqrt{2}$, the Tsirelson bound. CP violation rotates the correlations but does not dilute them; entanglement and CP structure are measured simultaneously from the same matrix.

Extraction from data.—In $\tau \rightarrow \pi\nu$ the pion is a perfect spin analyser ($\alpha_\pi = 1$) [22]: the polarimeter direction $\hat{h}$ in the tau rest frame is the pion direction, with $d\Gamma \propto 1 + \vec{s} \cdot \hat{h}$. The joint angular density is

$$p(\hat{h}^+, \hat{h}^- | C) = \frac{1}{(4\pi)^2} (1 - \hat{h}^{+\top} C \hat{h}^-), \quad (3)$$

where the sign reflects the opposite analysing-power convention for τ^+ and τ^- . The moment estimator is $\hat{C}_{ij} = -9 \langle \hat{h}_i^+ \hat{h}_j^- \rangle$, the factor 9 following from $\langle (\hat{h}_i)^2 \rangle = \frac{1}{3}$. The polarimeter acoplanarity angle ϕ (the azimuthal opening of $\hat{h}^+$ and $\hat{h}^-$ about $\hat{k}$) inherits the modulation

$$\frac{d\Gamma}{d\phi} \propto 1 - \frac{\pi^2}{16} \cos(\phi - 2\Delta), \quad (4)$$

where $\pi^2/16 \simeq 0.617$ combines $C_{nn} = C_{rr} = 1$ with $\langle \sin \theta \rangle^2 = (\pi/4)^2$ from each polar average [24, 25].

Estimators and their sampling properties.—A reliable nonlocality claim at small N requires care: the eigenvalue construction makes $\hat{m}_{12}$ a *noise-biased* statistic. Under the no-correlation null, $\hat{C} \rightarrow (3/\sqrt{N})G$ with G_{ij} i.i.d. standard normal, so $\hat{m}_{12}$ is $9/N$ times the sum of the two largest eigenvalues of a 3×3 Wishart matrix with three degrees of freedom:

$$E[\hat{m}_{12} | C = 0] = \frac{9 \langle \lambda_1 + \lambda_2 \rangle}{N} = \frac{78.5}{N}. \quad (5)$$

For $N = 50$ this “noise floor” is 1.6—above the Bell threshold $m_{12} = 1$ —so significance counting based on $\hat{m}_{12}$ alone is meaningless at small N . We therefore base all hypothesis tests on statistics *linear* in C (hence unbiased) or on exactly calibrated likelihood ratios:

(1) *CHSH score with fixed axes.* Choosing the measurement axes a priori from the SM prediction ($\hat{a} = \hat{n}$, $\hat{a}' = \hat{r}$, $\hat{b}, \hat{b}' = (\hat{n} \pm \hat{r})/\sqrt{2}$), the CHSH combination is $\langle S \rangle = \sqrt{2}(C_{nn} + C_{rr})$, estimated by the per-event quantity $s = -9\sqrt{2}(\hat{h}_n^+ \hat{h}_n^- + \hat{h}_r^+ \hat{h}_r^-)$ with $E[s] = 2\sqrt{2}$ and $\text{Var}[s] = 28$ under the SM (exactly), and $\text{Var}[s] = 36$ under the null. Local hidden-variable theories require $S \leq 2$ [3].

(2) *Concurrence witness.* The fidelity to the Bell state, $F = \langle \psi^+ | \rho | \psi^+ \rangle = (1 + C_{nn} + C_{rr} - C_{kk})/4$, bounds the concurrence from below, $\mathcal{C} \geq \mathcal{W} \equiv 2F - 1$ [14]; any separable state has $\mathcal{W} \leq 0$ while the SM gives $\mathcal{W} = 1$. The per-event estimator has $\text{Var}[w] = 9/2$ under the SM (exactly), so separability is rejected at 5σ with only $N \gtrsim 113$ events.

(3) *Optimal no-correlation test.* Against the hypothesis that the two decays are uncorrelated ($p_0 = (4\pi)^{-2}$), the Neyman–Pearson optimal statistic is the per-event log-likelihood ratio $t = \ln(1 - \hat{h}^{+\top} C_{\text{SM}} \hat{h}^-)$, for which we find the closed-form moments

$$\begin{aligned} E_{\text{SM}}[t] &= \ln 2 - \frac{1}{2}, & \text{Var}_{\text{SM}}[t] &= \frac{1}{4}, \\ E_0[t] &= \ln 2 - 1, & \text{Var}_0[t] &= 1. \end{aligned} \quad (6)$$

The expected significance for rejecting “no correlation” is therefore $Z = \sqrt{N}(E_{\text{SM}} - E_0)/\sqrt{\text{Var}_0} = \sqrt{N}/2$ exactly. The test is calibrated without any Gaussian assumption by *permutation*: re-pairing τ^+ and τ^- hemispheres across events realises the null exactly in data, preserving all single-tau distributions and detector effects while destroying the correlations.

Causal-speed distribution.—From the reconstructed decay four-positions we form $\Delta s^2 = c^2 \Delta t^2 - \Delta r^2$ (invariant) and the minimum signal speed required to connect the decays, $v_{\text{sig}} = \Delta r/(c \Delta t)$, evaluated in the Higgs rest frame. There the taus are exactly back-to-back with equal speed β , and decay lengths $L_{\pm}$ are i.i.d. exponential. Since $(L_+ - L_-)/(L_+ + L_-)$ is exactly uniform on $[-1, 1]$ for exponential variables,

$$v_{\text{sig}} = \beta \frac{L_+ + L_-}{|L_+ - L_-|} \implies P(v_{\text{sig}} > v) = \frac{\beta}{v}, \quad v \geq \beta, \quad (7)$$

an exact Pareto law: half of all events require superluminal connections faster than $2c$, and a fraction $1/v$ faster than vc . Under a finite-speed causal model [7, 8] in which correlations are carried by an influence propagating at v_ψ in the Higgs frame, all events with $v_{\text{sig}} > v_\psi$ must be uncorrelated. Combining Eq. (7) with $Z = \sqrt{N}/2$, the median 95% C.L. exclusion reach is

$$v_\psi^{95} = \frac{N\beta}{4 z_{95}^2} \simeq \frac{N}{15.4}, \quad (8)$$

linear in the event count N .

Simulation and reconstruction.—We analyse simulated $e^+e^- \rightarrow ZH \rightarrow \mu^+\mu^-\tau^+\tau^-$ events at $\sqrt{s} = 240$ GeV with both taus decaying to $\pi\nu$. From 1003 generated $\mu^+\mu^-\tau^+\tau^-$ events, 90 satisfy the $\pi \times \pi$ requirement and are fully reconstructed; this corresponds to $\sim 15 \text{ ab}^{-1}$ in this single channel, and we use Eq. (8) to project to realistic multi-channel datasets below. The Higgs four-momentum is $p_H = p_{\text{beam}} - p_{\mu^+} - p_{\mu^-}$, exploiting the known beam energy. The tau direction and decay vertex follow from the impact-parameter method [26]: the pion track and the production vertex define a plane containing the tau momentum; the opening angle α fixes the decay length $L = d/\sin \alpha$ and the τ mass constraint fixes the momentum, with the two-fold ambiguity resolved by the missing-momentum constraint $p_H - p_{\pi^+} - p_{\pi^-}$. Tau energies are then set by the Higgs-frame constraint $E_\tau = m_H/2$. Vertex-quality requirements of 3σ impact-parameter significance and a fiducial decay length $L <$

100 mm on both taus (the impact-parameter geometry is only valid for decays inside the vertexing volume) retain 49 events for vertex-dependent observables; the spin observables use all 90. Detector response is modelled by Gaussian smearing of ILD-like track parameters ($\sigma_{d_0} = \sqrt{5^2 + (10/p \sin^{3/2} \theta)^2} \mu\text{m}$, $\sigma_{\theta, \phi} = 0.1 \text{ mrad}$, $\sigma_{1/p_T} \simeq 2 \times 10^{-5} \text{ GeV}^{-1}$) [29]; the typical tau impact parameter at $\gamma \approx 35$ is comparable to σ_{d_0} , so smearing broadens v_{sig} by a measured 16% fractional resolution while leaving the C_{ij} extraction essentially intact.

Spin correlations.—Figure 1 shows the reconstructed correlation matrix, consistent with Eq. (2): the diagonal is $(+1.01, +0.98, -0.88) \pm 0.3$, off-diagonals consistent with zero. The unbiased observables give

$$S = 2.81 \pm 0.58, \quad \mathcal{W} = 0.94 \pm 0.24, \quad (9)$$

both consistent with the SM expectations $2\sqrt{2}$ and 1, and with truth-level validation ($S = 2.81 \pm 0.58$, $\mathcal{W} = 0.91 \pm 0.24$): after the audit-driven refinement of the direction-scan granularity the reconstruction adds no measurable dilution in the tracking-limited baseline. Separability ($\mathcal{W} \leq 0$) is rejected at 3.8σ : the tau pairs are certifiably entangled. The raw Horodecki parameter, $\hat{m}_{12} = 2.57 \pm 1.03$, illustrates Eq. (5): its noise floor at $N = 90$ is 0.87, and the bootstrap-bias-corrected value is 1.54; we therefore do not base significance claims on it. The optimal LR test rejects “no correlation” with permutation $p < 10^{-4}$ ($Z = 4.3$, against the ideal expectation $\sqrt{90/2} = 4.7$). The acoplanarity modulation, $B = -0.67 \pm 0.13$ at truth level and -0.58 ± 0.16 after reconstruction, confirms the exact prediction $-\pi^2/16 = -0.617$ of Eq. (4) (a frequently quoted value of -0.5 is not the SM expectation for this channel). With full detector smearing the central values shift by less than one standard deviation ($S = 2.44 \pm 0.62$, $\mathcal{W} = 0.81 \pm 0.27$, LR $Z = 3.3$).

Spacetime-resolved entanglement.—Figure 2 shows m_{12} (bias-corrected) and the witness $\mathcal{W}$ versus the signed interval $\text{sgn}(\Delta s^2) \sqrt{|\Delta s^2|}$. All 49 vertex-selected events are spacelike separated, a consequence of the back-to-back topology, the ultrarelativistic boost, and the vertex-quality requirement (timelike configurations require one near-prompt decay, which fails the impact-parameter cut). At truth level the witness is consistent with $\mathcal{W} = 1$ in every spacelike bin (open markers): entanglement persists undiminished across spacelike separations, and any residual trend in the reconstructed values reflects decay-length-dependent resolution, not decoherence.

Finite-speed exclusion.—For each hypothesis v_ψ we select the events with $v_{\text{sig}} > v_\psi$ and apply the permutation-calibrated LR test. Figure 3 shows the result: the no-correlation hypothesis is rejected at 95% C.L. out to $v_\psi \approx 5c$ (the last populated point, at $10c$ with six events, also exceeds the threshold), consistent within the expected fluctuations with the median analytic reach

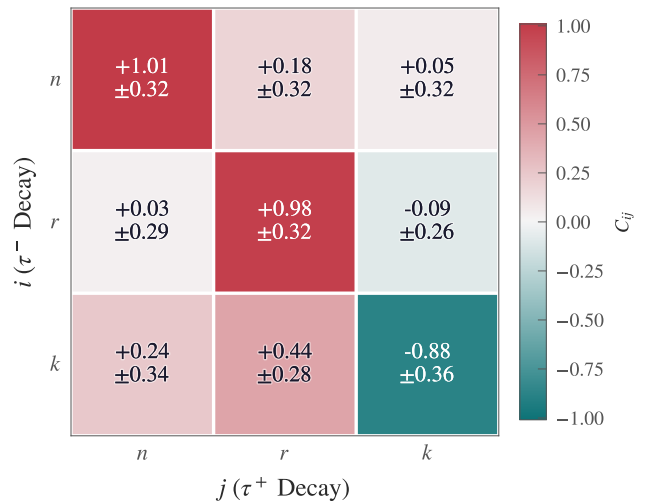


FIG. 1. Reconstructed spin correlation matrix C_{ij} in the $\{n, r, k\}$ basis from 90 $\tau \rightarrow \pi\nu$ pairs. Entries show the central value $\pm$ bootstrap uncertainty. The exact SM prediction, Eq. (2), is $C = \text{diag}(+1, +1, -1)$ for any β .

$v_\psi^{95} = 3.2c$ from Eq. (8) with $N = 49$. The surviving event counts follow the Pareto law of Eq. (7) (Fig. 3, bottom; observed median $v_{\text{sig}} = 3.0$, predicted $2\beta \simeq 2$, with the difference attributable to resolution and the vertex-quality selection, which preferentially removes low- v_{sig} configurations).

Projections.—Equation (8), validated above on simulation, converts event counts directly into physics reach. At the full ILC program (2 ab^{-1} at $\sqrt{s} = 250 \text{ GeV}$ [28]), about 3.0×10^4 $H \rightarrow \tau^+\tau^-$ decays are produced. Including $Z \rightarrow q\bar{q}$ and $Z \rightarrow ee, \mu^+\mu^-$ (76% of Z decays), $N \simeq 3 \times 10^2$ $\pi \times \pi$ pairs are expected, giving $v_\psi^{95} \approx 20c$; entanglement certification at 5σ needs only ~ 110 such events. The $\tau \rightarrow \rho\nu$ channels would increase the yield tenfold: the $\rho\nu$ polarimeter vector $H^\mu = 2(qN)q^\mu - q^2N^\mu$ (with $q = p_{\pi^\pm} - p_{\pi^0}$) is exactly lightlike and carries unit analysing power when the neutrino is reconstructed [23], and in our simulation it closes on the SM at truth level; however, the kinematically inferred neutrino dilutes the azimuthal correlations substantially ($\langle \cos \delta\phi \rangle \simeq 0.5\text{--}0.7$), so exploiting these channels—and a reach beyond 10^2c , with a 5σ Bell-inequality violation ($N \gtrsim 10^3$ at the ideal $S = 2\sqrt{2}$)—requires a global kinematic fit of the tau pair, which we leave to future work.

Discussion.—Dedicated photon experiments bound the speed of a hypothetical “spooky action” to $\gtrsim 10^4c$ for particular preferred frames [7], far above the reach derived here. The measurement we propose is nevertheless qualitatively new in three respects. First, the entangled system consists of *massive fermions* produced by the *Yukawa interaction*—a sector in which nonlocality has never been tested; given the weak sector’s record of broken symmetries, this is a test worth performing,

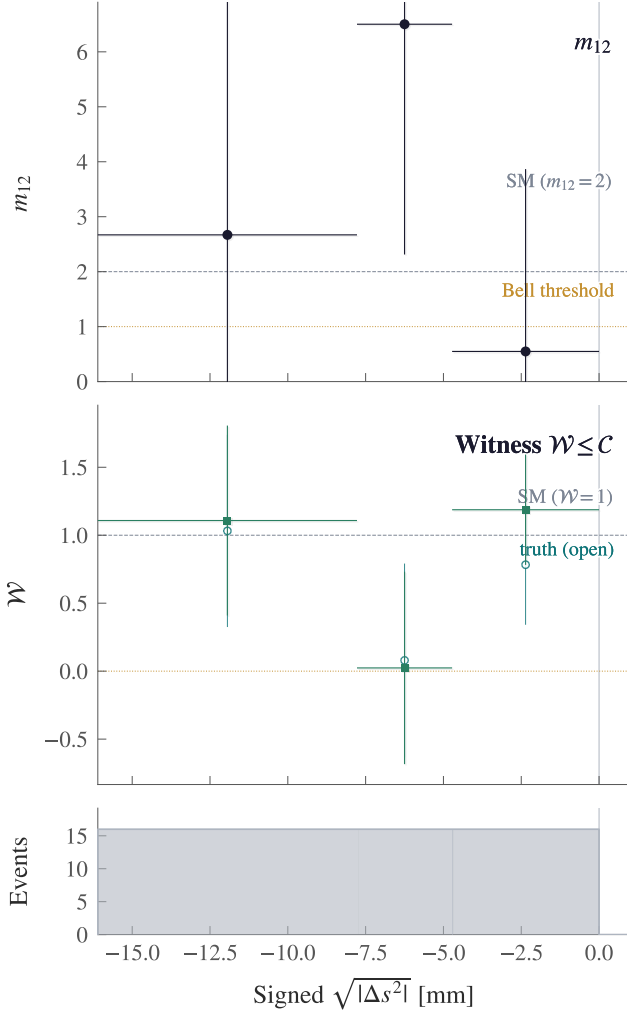


FIG. 2. Bias-corrected m_{12} (top) and concurrence witness $\mathcal{W}$ (middle) versus the signed spacetime interval between the two tau decay vertices; negative values are spacelike. Open markers show the truth-level witness, consistent with the SM value $\mathcal{W} = 1$ in every bin; differences between filled (reconstructed) and open markers reflect resolution, not decoherence. Bottom: event counts. Dashed lines mark the SM predictions and the Bell ($m_{12} = 1$) and separability ($\mathcal{W} = 0$) thresholds.

not assuming. Second, the spacelike separation is not engineered but occurs naturally at the micrometre-to-millimetre scale, probing causal structure at microscopic distances and at a relativistic boost ($\gamma \approx 35$) inaccessible to optical experiments. Third, the same dataset simultaneously determines the CP structure of the $H\tau\tau$ coupling through the acoplanarity phase 2Δ in Eq. (4): nonlocality and CP violation are measured from one correlation matrix. As in all decay-based tests, the “measurement settings” are not free choices, so the freedom-of-choice loophole remains open [27]; the measurement excludes the class of finite-speed causal models defined above rather than all local theories.

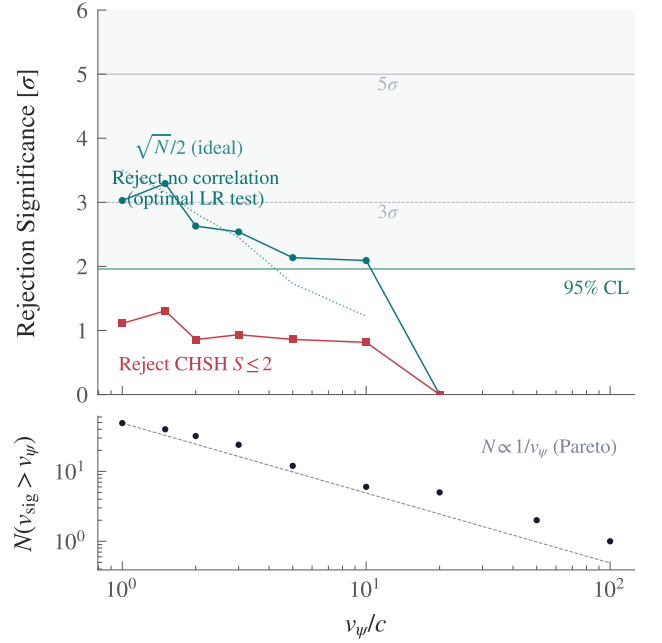


FIG. 3. Significance for rejecting the finite-speed causal model versus the hypothesised speed v_ψ . Circles: optimal LR test against “no correlation” (permutation-calibrated); the dotted curve is the ideal expectation $\sqrt{N(v_\psi)}/2$ from Eq. (6). Squares: CHSH $S \leq 2$ rejection. Horizontal lines mark 95% C.L., 3σ and 5σ . Bottom: surviving event counts, following the exact Pareto law $P(v_{\text{sig}} > v_\psi) = \beta/v_\psi$ (dashed).

In summary, we have presented the complete statistical framework for a spacetime-resolved entanglement measurement in $H \rightarrow \tau^+\tau^-$ at an e^+e^- Higgs factory: exact SM expectations [Eqs. (2), (4)], unbiased estimators with closed-form variances, an optimal permutation-calibrated test [Eq. (6)], and an analytic, simulation-validated sensitivity law [Eqs. (7), (8)]. The measurement certifies entanglement between spacelike-separated, Yukawa-produced massive fermions and excludes finite-speed causal models up to speeds growing linearly with the dataset—a unique capability of future Higgs factories.

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